

Thirty Years of Global Disease Outbreaks: What the Data Actually Show

A country-by-country look at outbreak trends from 1996 to 2026, the surveillance shift of 2020, and the recent rise in non-COVID disease events

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Abstract

For years, global outbreak trackers have shown a steady climb in reported infectious disease events, and it is tempting to read that as proof that outbreaks themselves are becoming more common. But is that really what is happening, or have we simply gotten better at spotting and reporting them? We looked at 30 years of global outbreak data, from 1996 to 2026, covering 3,566 recorded events, to try to answer that.

We built a full country-by-country panel covering 236 countries and territories over 31 years, which works out to 7,316 country-year observations in total. Years with zero outbreaks were counted too, not just the years when something happened, since leaving them out would bias the picture. Because the data was noticeably overdispersed (variance running about 1.4 times higher than the mean), we used negative binomial regression instead of a plain Poisson model. On top of that, we used an interrupted time-series approach to test whether outbreak counts followed one smooth trend line, or whether something changed abruptly around 2020. We then reran the whole analysis after removing every COVID-19 record, to check whether the results were just about COVID reporting or reflected something broader.

Outbreaks are heavily concentrated in a handful of countries, led by the Democratic Republic of the Congo (80 outbreaks over the period) and Nigeria (61), both in Central and West Africa. If you fit one straight trend line across the full 30 years, outbreaks appear to be growing 7.68% a year, which would imply more than a sixfold increase over 25 years. That number turns out to be misleading. Once the model accounts for a structural break in 2020, the story changes: outbreak counts were essentially flat before 2020, then jumped 5.6-fold that year, before easing off by roughly 3.8% a year afterward as emergency COVID-era reporting wound down. The more telling result shows up once COVID is removed from the data entirely. Non-COVID outbreaks were also flat before 2020, jumped 43.7% that year, and then kept climbing afterward by about 13.1% every year through 2026.

What looks like three decades of steady growth in global outbreaks turns out to be mostly a story about when and how well we started counting. That said, the years since 2020 tell a real and separate story: non-COVID disease outbreaks are genuinely on the rise, likely reflecting both the expanded lab and testing capacity built during the pandemic and a genuine resurgence of diseases such as measles, cholera, and meningitis following years of disrupted vaccination programs and strained health systems.

1. Why This Matters

Outbreaks of infectious disease are one of the clearer threats to public health, economic stability, and the resilience of health systems worldwide. Over the last few decades, global health registries have recorded more and more outbreak events each year, and that raw trend gets cited often as evidence of a worsening disease landscape. But raw counts from a registry like this are tricky to interpret. A rise in recorded outbreaks could mean diseases are genuinely spreading more, driven by things like climate change, deforestation, intensive agriculture, and more cross-border travel. Or it could simply mean we are watching more closely: better lab testing, wider genomic sequencing, and stronger compliance with international health reporting rules all make it more likely that an outbreak gets noticed and logged in the first place.

Most reports on this kind of data lean on simple descriptive summaries or a single trend line fit across the whole time period. That approach glosses over the fact that surveillance itself is not static, and it completely misses the enormous mobilization of testing and reporting infrastructure that came with the COVID-19 pandemic in 2020. Treating three decades of very different surveillance conditions as one smooth trend risks badly overstating how fast baseline disease risk is actually rising, and it hides what is going on with specific pathogens. On top of that, outbreak data is naturally lopsided: most countries report zero major outbreaks in any given year, while a small number of hotspots see repeated events across several diseases at once.

To work through these issues, this analysis uses 30 years of outbreak records (1996 to 2026, 3,566 events in total) and a negative binomial interrupted time-series model to do three things: map out which countries and pathogens carry the heaviest burden, test whether the historical trend is a smooth line or an abrupt shift, and check whether the pattern since 2020 is just a COVID reporting artifact or a genuine, ongoing rise in other diseases.

2. How We Did This

2.1 Building the panel

We started with 3,566 validated global outbreak records collected between January 1, 1996 and mid-2026. To model this properly and avoid the bias that comes from only looking at years where something happened, we reshaped the raw event list into a balanced country-year panel: every one of 236 countries and territories gets one row for each of the 31 years in the study, including a count of zero for any country-year with no recorded outbreak. That gives 7,316 country-year observations in total.

2.2 Choosing the right model

A standard Poisson regression assumes the variance of outbreak counts equals the mean. That assumption did not hold here: the mean was 0.487 outbreaks per country-year, but the variance was 0.690, a ratio of about 1.41. Ignoring that kind of overdispersion tends to understate uncertainty and can make effects look more significant than they really are. We used negative binomial regression

instead, which allows the variance to exceed the mean, and compared model fit using the Akaike Information Criterion (AIC), where a lower number indicates a better-fitting model.

2.3 Three models, one question

Model 1 (straight trend): a simple model with outbreak count as a function of calendar year, the kind of model most reports default to.

Model 2 (structural break): an interrupted time-series model that lets the data show a possible step change in 2020 (an immediate jump up or down) plus a possible change in slope from 2020 onward, on top of whatever the pre-2020 trend was doing.

Model 3 (non-COVID only): the same structural-break model, rerun after removing all 1,214 COVID-19 records, leaving 7,098 residual degrees of freedom. This tells us whether any post-2020 shift is really just COVID being logged as an outbreak, or whether something bigger is going on across other diseases too.

Results are reported as incidence rate ratios (IRRs), which show how much more (or less) likely an outbreak is in a given year compared to the baseline, along with 95% confidence intervals. An IRR of 1.05, for example, means a 5% increase; an IRR of 0.95 means a 5% decrease. All analysis was done in R (version 4.6.1) using the MASS package.

3. What We Found

3.1 Where outbreaks happen, and what causes them

Outbreak reporting is far from evenly spread across the globe. Five countries account for a disproportionate share of all recorded events: the Democratic Republic of the Congo (80), Nigeria (61), the United States (51), China (49), and Brazil (47).

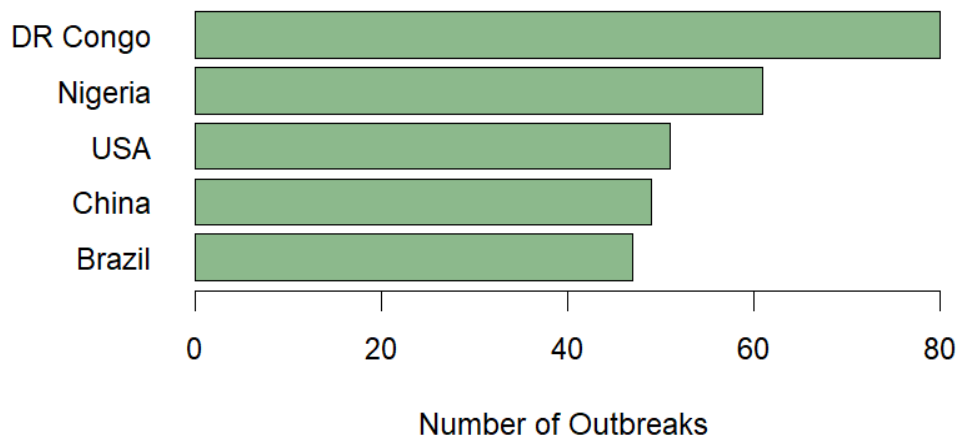


Figure 1. The five countries with the most recorded outbreaks between 1996 and 2026. The Democratic Republic of the Congo and Nigeria, both in Central and West Africa, lead by a wide margin.

The Democratic Republic of the Congo hit a 30-year high of 10 outbreaks in a single year, 2024. That peak was not one large epidemic; it was seven different diseases circulating at the same time, including Mpox (Clade I), Ebola, cholera, and measles.

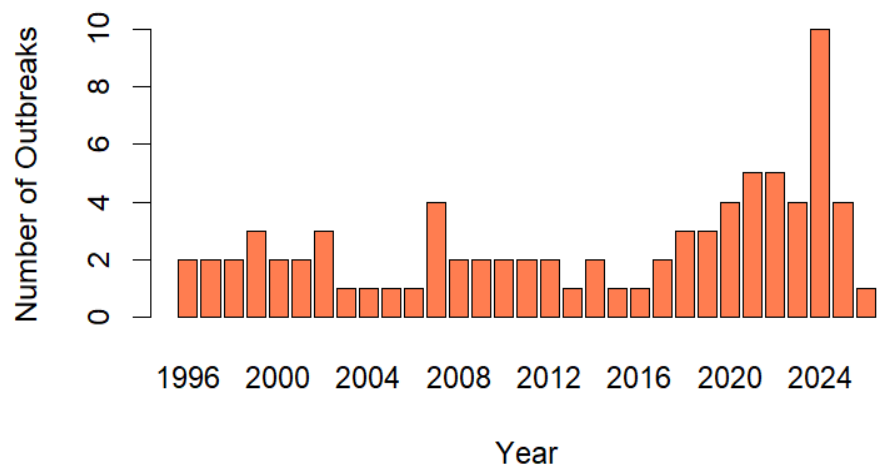


Figure 2. Outbreaks recorded in the Democratic Republic of the Congo by year. Activity was low and fairly flat for most of the study period, then rose sharply after 2019 and peaked at 10 outbreaks in 2024.

Nigeria's 61 recorded outbreaks are dominated by hemorrhagic fevers (18 events), followed by bacterial diseases (9), intestinal or waterborne infections (8), other emerging pathogens (7), and viral infections of the central nervous system (7). Reporting there clustered around 2017 and 2023, years marked by overlapping Lassa fever, yellow fever, and meningococcal meningitis activity.

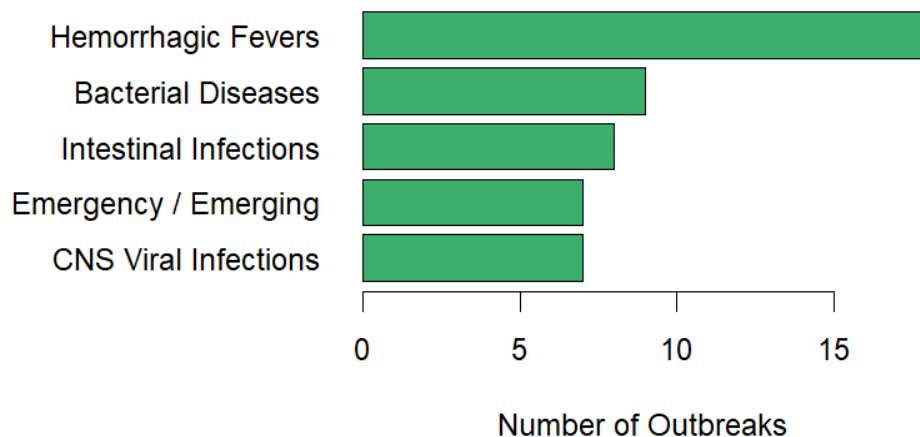


Figure 3. Types of outbreaks recorded in Nigeria. Hemorrhagic fevers such as Lassa fever make up close to a third of the country's total.

Looking across the whole dataset, respiratory viruses top the global list, led by COVID-19 (1,214 events) and pandemic or zoonotic influenza (554). After that come cholera (245), dengue (209), yellow fever (159), polio (125), Mpox (108), measles (99), meningococcal meningitis (96), and MERS-CoV (76).

3.2 Steady climb, or sudden jump?

This is where the choice of model changes the conclusion entirely. Fit a single trend line across all 30 years, and global outbreaks look like they grew 7.68% a year (95% CI: 7.2% to 8.2%), which compounds to roughly a sixfold increase between 1996 and 2021. That is the number a simple year-over-year trend chart would suggest, and it is the kind of figure that ends up quoted as proof of an accelerating global disease crisis.

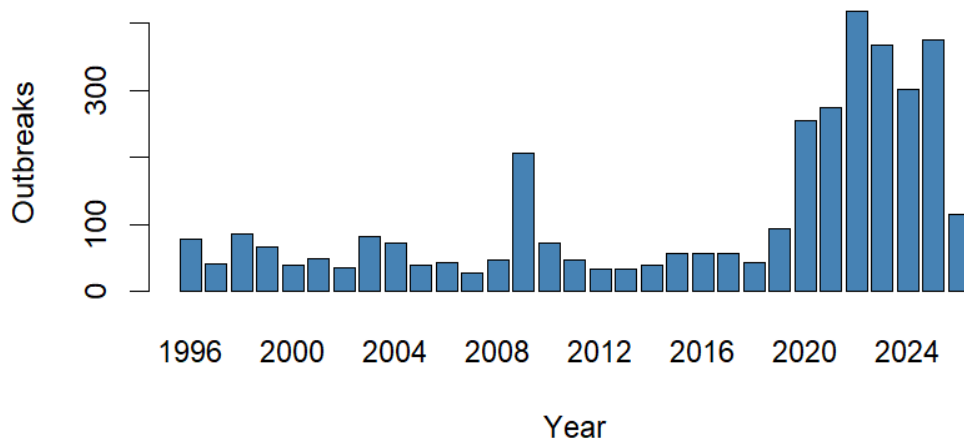


Figure 4. Total recorded outbreaks worldwide by year. The dramatic jump starting in 2019 to 2020, and the partial pullback after 2022, is exactly the kind of structural shift that a single trend line glosses over.

Letting the model account for a break in 2020 tells a very different story. Allowing for that break improved the model fit substantially (AIC dropped by 725 points, a large improvement). Under this model, the pre-2020 trend essentially disappears: there was no meaningful year-over-year change in global outbreak counts between 1996 and 2019 (IRR = 0.999, 95% CI: 0.992 to 1.006, not statistically significant). Instead, counts stayed roughly flat, in the range of 50 to 60 events a year, right up until 2020, when reporting jumped 5.6-fold almost overnight (95% CI: 4.95 to 6.44). After that initial surge, aggregate counts actually declined by about 3.8% a year (95% CI: 2.2 to 6.0% decline) as emergency pandemic-era tracking systems were gradually scaled back.

3.3 Taking COVID out of the picture

The obvious next question is whether that 2020 jump was just COVID-19 itself being counted as an outbreak. So we removed all 1,214 COVID-19 records and reran the model on everything else. Two

things stood out. First, non-COVID outbreaks were just as flat before 2020 as the overall data (IRR = 0.999, not statistically significant), which confirms the pre-2020 baseline really was stable. Second, and more notably, non-COVID outbreaks still jumped 43.7% in 2020 (95% CI: 17.3% to 75.8%), and instead of easing off afterward like the aggregate numbers did, they kept accelerating, climbing by about 13.1% every year from 2020 through 2026 (95% CI: 8.2% to 18.3%).

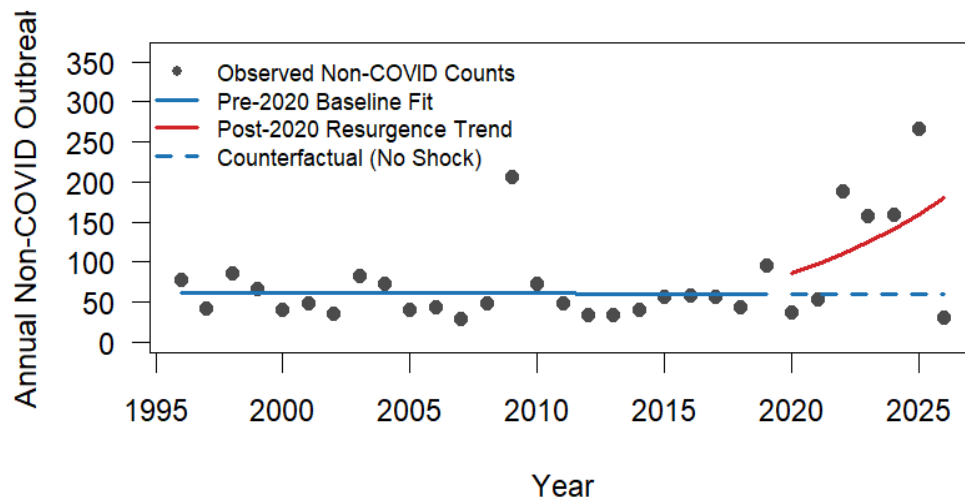


Figure 5. Non-COVID outbreaks worldwide, 1996 to 2026. The flat pre-2020 baseline (blue) is unchanged by removing COVID cases, but the years afterward (red) climb well above where the old baseline would have predicted (dashed line), reaching over 250 events by 2025.

So the picture that emerges is not one where COVID caused a temporary spike that is now fading. It is one where COVID coincided with, and likely helped trigger, a lasting rise in how often other diseases are being detected and reported, and that rise is still gaining pace.

Table 1. Consolidated three-model negative binomial regression comparison

Model	Empirical coefficient(s)	Dispersion parameter (theta)	AIC
1. Straight trend	beta_year = 0.0740 (IRR 1.0768)	4.741	12,633.1
2. 2020 structural break	beta_pre = -0.0010; beta_step = 1.7228; beta_post = -0.0387	28.070	11,908.0
3. Non-COVID structural break	beta_pre = -0.0010; beta_step = 0.3626; beta_post = 0.1231	1.132	10,378.0

4. What This Means

4.1 The headline finding

The common assumption, that global infectious disease outbreaks have been climbing steadily and exponentially for three decades, does not hold up once the data is modeled properly. Once the panel accounts for zero-outbreak years and the true shape of the data, outbreak reporting was essentially flat from 1996 through 2019. The apparent 7.68% annual growth that shows up in a simple trend line is largely an artifact of jamming two very different surveillance eras, before and after 2020, into a single straight line.

What is real is the break itself. 2020 marks a genuine turning point in how the world tracks disease. While the aggregate numbers pulled back after the initial COVID surge, once COVID is excluded, other diseases kept getting reported more often, at a compounding rate of about 13.1% a year.

4.2 Why non-COVID outbreaks kept climbing

Two things are probably driving this, and they likely reinforce each other. The first is capacity: the massive expansion of PCR testing, genomic sequencing, and digital disease-reporting systems built for COVID did not disappear once the emergency passed. That infrastructure lowered the bar for detecting other diseases too, so illnesses that used to go undiagnosed, or get lumped together as a vague fever, are now being identified and logged in international registries.

The second is a genuine rebound in disease activity. Routine childhood vaccination was badly disrupted during 2020 to 2022, leaving large pockets of unprotected people and setting the stage for measles, polio, and meningitis outbreaks. At the same time, climate pressures and strained health systems have made it easier for vector-borne and waterborne diseases like dengue and cholera to spread. In the hardest-hit countries, like DR Congo and Nigeria, several of these threats are showing up together: hemorrhagic fevers, Mpox, and bacterial infections all circulating in health systems that are already under pressure.

4.3 What this means for policy

This matters for how global health funding and preparedness planning get done. Projections that assume outbreak risk will keep growing along the old, unadjusted trend line will overstate the pre-2020 baseline and misjudge how urgent the response needs to be. At the same time, the sustained 13.1% annual rise in non-COVID outbreaks is a real signal that the disease landscape today is more varied, and arguably riskier, than it was before the pandemic. Dismantling the lab and surveillance capacity built during COVID, simply because the emergency has passed, would mean giving up the very tools that are now catching these co-circulating outbreaks in the first place.

4.4 Caveats worth keeping in mind

A few limitations are worth flagging. Reporting bias by country is real: wealthier nations and countries under closer international scrutiny tend to log more events than lower-resource settings with less

diagnostic capacity, regardless of actual disease burden. The 2026 figures are incomplete, since the year was not yet over when the data was pulled, so that final data point should be read with caution. And while we removed COVID cases specifically to isolate non-COVID trends, the pandemic's broader disruption to everyday healthcare cannot be fully separated from the numbers we see for other diseases.

5. Bottom Line

Three decades of rising global outbreak numbers turn out to be mostly a story about surveillance catching up, not disease spreading faster, at least up until 2020. What happened after 2020 is different: non-COVID infectious disease events have been genuinely accelerating, at close to 13% a year, driven by a combination of better detection and a real resurgence of endemic diseases following pandemic-era disruptions. Keeping surveillance and lab infrastructure funded, and directing more resources toward chronic hotspots in Central and West Africa, looks like a sound investment based on what this data shows, well before these trends turn into the next international health emergency.